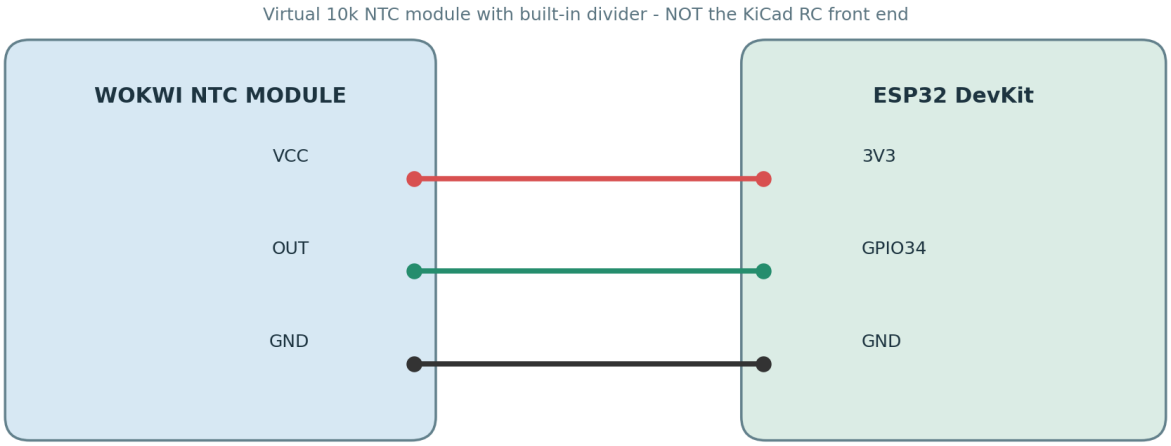
# Wokwi - Verified ESP32 Firmware Bench

User-executed Wokwi runs; virtual sensor module directly connected to the ESP32 ADC input.

## 01 / VIRTUAL WIRING




| Wokwi NTC module | ESP32 DevKit | Purpose |
|---|---|---|
| VCC | 3V3 | 3.3 V virtual supply |
| GND | GND.1 | Ground reference |
| OUT | D34 / GPIO34 | ADC1 analog input |

## 02 / WHAT THIS SIMULATION PROVES

The supplied sketch.ino boots in Wokwi and reports ms,mv,temp_c,status at nominal 200 ms intervals. The thermistor temperature slider drove measured changes in the ADC voltage and calculated temperature. The experiment exercises virtual sensor acquisition, Beta-model conversion and state transitions.

Evidence boundary. Wokwi uses an internal 10 kOhm divider and does not reproduce the KiCad analog circuit with R1 = 9.82 kOhm, R2 = 990 Ohm and C1 = 100 nF. LTspice covers that electrical model. All Wokwi values are virtual, not physical calibration or fabricated-PCB results.

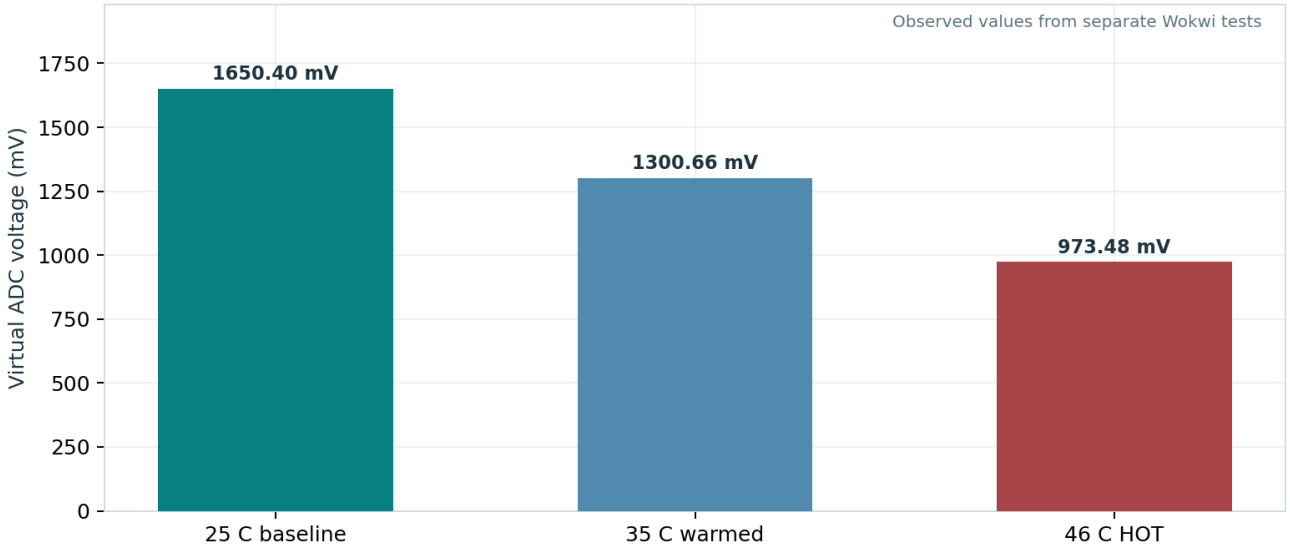
# 25, 35 and 46 C - observed response

Recorded values from user-shared Wokwi serial monitor; distinct test sessions are not spliced into a continuous time series.

## 01 / SELECTED ACTUAL VIRTUAL OBSERVATIONS

| Temperature setpoint | Serial-reported value | Status |
|---|---|---|
| 25 C baseline | 1650.40 mV / 24.989 C | NORMAL |
| 35 C stabilized | 1300.66 mV / 35.000 C | NORMAL |
| 46 C hold | 973.48 mV / 45.987 C | HOT |

## 02 / LOWER VOLTAGE AT HIGHER TEMPERATURE




## 03 / CONTROL AND REPORTING VALIDATION

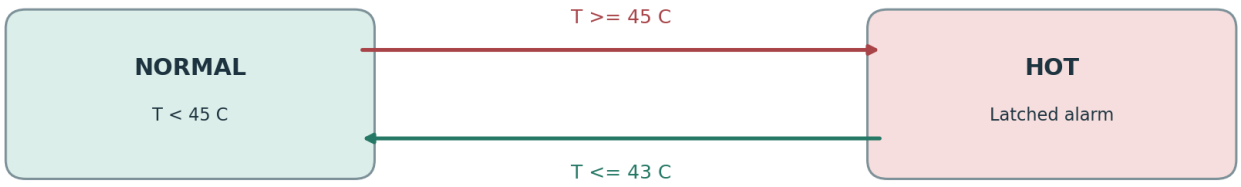
At 25 C the sketch reported 24.989 C / NORMAL. In the 35 C run, it changed from 24.989 C to 34.902 C and then 35.000 C. In the 46 C run, the alarm switched to HOT at 45.386 C and remained HOT during the 45.987 C hold. Timestamps incremented by 200 ms across logged settled intervals. These values are observed simulator outputs, not predicted values.

Source: `Wokwi_Observed_Log_Excerpts.csv` and original screenshots. CSV contains selected exact logged rows, not the entire serial session.

# HOT Alarm and Hysteresis - observed

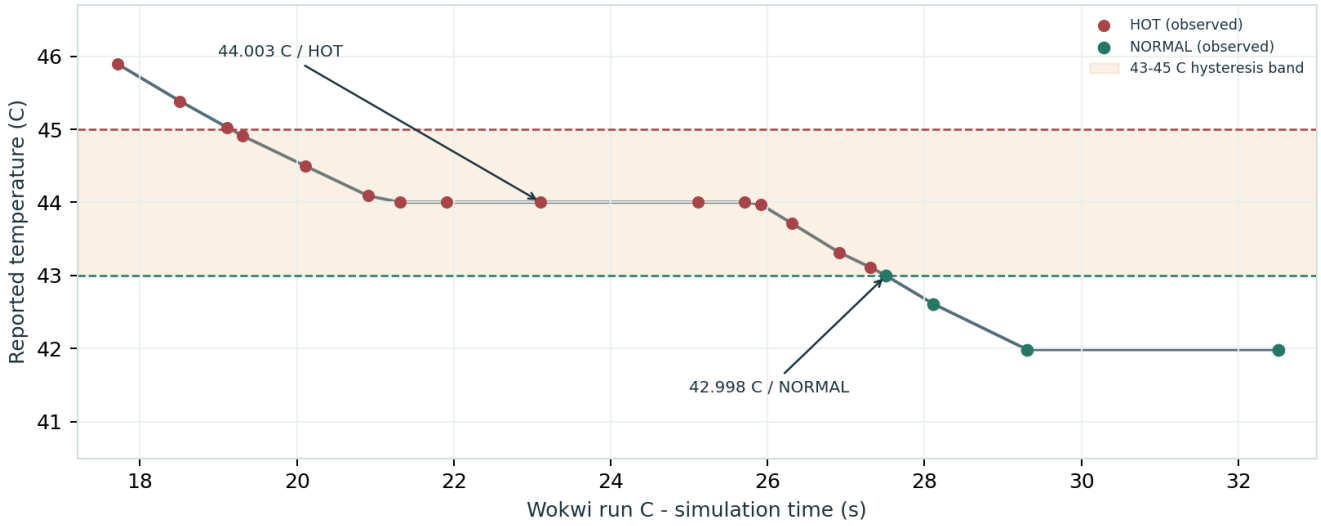
The 2 C hysteresis band was explicitly exercised in Wokwi on the cooling path.

## 01 / FIRMWARE STATE LOGIC




OUT_OF_RANGE and voltage faults are distinct status outputs; current firmware clears the HOT latch on faults.

## 02 / 44 C HOLD AND 43 C RESET

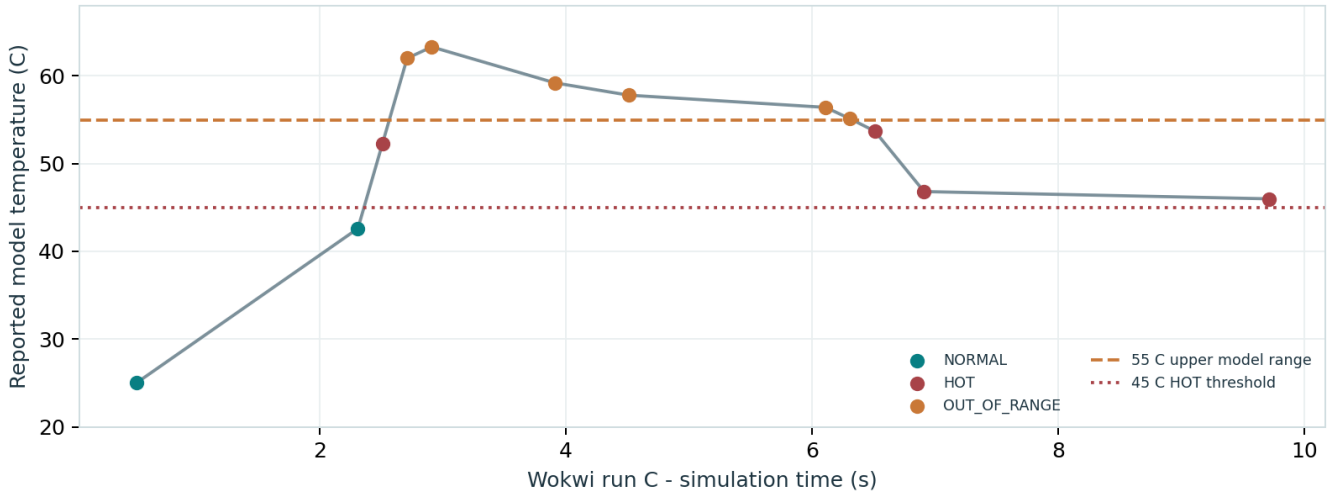



| Observed temp | Status | Interpretation |
|---|---|---|
| 45.386 C | HOT | Over 45 C turns the alarm on |
| 44.003 C | HOT | Within hysteresis band - stays on |
| 43.112 C | HOT | Just above reset - stays on |
| 42.998 C | NORMAL | At or below 43 C - resets |

# Out-of-Range Event - captured, not hidden

A rapid setpoint change caused simulated temperature excursions above the configured 55 C software range.

## 01 / OBSERVED OVER-RANGE BEHAVIOR




| Time (ms) | Reported temperature | Reported state |
|---|---|---|
| 2510 | 52.301 C | HOT |
| 2710 | 62.007 C | OUT_OF_RANGE |
| 2910 | 63.304 C | OUT_OF_RANGE |
| 6510 | 53.696 C | HOT |

## 02 / CORRECT INTERPRETATION

The OUT_OF_RANGE branch fired as coded. However, the current Wokwi sketch clears the HOT latch when an over-range or voltage-fault condition occurs. A production alarm should specify how faults affect operator warnings and whether an alarm must remain latched after sensor recovery. This is a documented improvement opportunity, not a fully verified safety feature.

No separate low-voltage or high-voltage fault-injection test has been evidenced; only OUT_OF_RANGE is observed.

# Cross-Tool Verification Summary

What is supported by actual source, what is only simulated, and what remains unverified.

## 01 / TOOL-TO-EVIDENCE MAP

| Evidence source | Supported claim | Boundary |
|---|---|---|
| Physical ESP32 + DMM | Voltage/thermal response; 641-sample steady-state SD = 0.0412 C | Breadboard only; no reference temperature accuracy |
| KiCad 10 native CAD | Editable schematic; routed 50 x 35 mm PCB; 3D model; reported 0 active DRC errors | Unfabricated; five ignored DRC categories and fit checks |
| LTspice user-run files | 1.5791 V simulated at 26 C; 267.7 Hz nominal model filter cutoff | Nominal NTC model, not calibration |
| Wokwi user-run logs | 25/35/46 C response, HOT activation, 44 C hold, below-43 C reset | Virtual NTC internal divider, not analog PCB |
| Python / firmware files | Serial parsing, dashboard, analysis code, tests | Not all hardware firmware binaries traceable |

## 02 / ENGINEERING CONCLUSION

The project now demonstrates a traceable engineering chain: analytical modeling -> circuit simulation -> embedded virtual testing -> physical breadboard measurements -> PCB CAD documentation. These are distinct forms of evidence. The hardware temperature-accuracy claim and operation of a manufactured PCB are deliberately excluded.

# Reproduction, Evidence Files and Handoff

Source files, key tests, open limitations, and a short interview explanation.

## 01 / EXACT SOURCE FILES

| Folder | Contents |
|---|---|
| 02_KiCad_Source | Editable .kicad_pro, .kicad_sch and .kicad_pcb |
| 03_CAD_Visuals | Schematic PDF, PCB copper/outline plots and 3D images |
| 05_Hardware_CSV / 06_Analysis | Four measured ESP32 captures + recomputed graphs |
| 09_LTspice_Simulation | Original .cir, .raw, .log + Bode and sweep graphs |
| 10_Wokwi_Simulation | Verified sketch.ino, diagram.json, selected logs, figures, screenshots |

## 02 / REPEAT THE WOKWI TEST

Open a new ESP32 Wokwi project. Paste the supplied sketch.ino and diagram.json; start the simulator. Confirm 25 C / NORMAL; change to 35 C / NORMAL; move to 46 C / HOT. Cool to 44 C and verify the alarm remains HOT; cool below 43 C and verify NORMAL. Finally return to 25 C. Store any future full serial export alongside the selected rows included here.

## 03 / SHORT INTERVIEW EXPLANATION

"I designed and tested an ESP32 NTC-sensing prototype. LTspice let me predict divider and RC-filter behavior, while Wokwi let me exercise the virtual firmware at 25, 35 and 46 degrees. I verified the HOT latch remained active at 44 degrees and reset near 43 degrees during cooling. I also documented a simulated out-of-range event. The real breadboard provided the separate measurement dataset; the custom KiCad PCB remains an unfabricated design."

## 04 / NEXT REAL VALIDATION

Reference thermometer, ADC calibration, actual PCB fabrication/assembly, independent hardware fault tests, and mechanical screw/head-clearance checks.